

HDOC-14

Question:

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Check whether all its digits are unique.
- Find the frequency of each digit from `0` to `9`.

Task1: Algorithm:

Step 1: Start.

Step 2: Declare variables:

- num, temp as long long int
- choice, digit, i, unique as int
- freq[10] as an integer array

Step 3: Input a positive integer num.

Step 4: If num <= 0, display "Invalid input" and stop.

Step 5: Display the menu:

1. Check whether all digits are unique
2. Find frequency of each digit

Step 6: Input choice.

Step 7: Copy the number into temp and initialize all elements of freq[10] to 0.

Step 8: Use switch(choice).

Case 1: Check whether all digits are unique

1. Set unique = 1.
2. Repeat while temp > 0:
 - Find last digit: digit = temp % 10
 - Increase its frequency: freq[digit]++
 - If freq[digit] > 1, set unique = 0 and exit the loop.
 - Remove last digit: temp = temp / 10
3. If unique == 1, display "Unique".
4. Otherwise, display "Not Unique".

Case 2: Find frequency of digits

1. Repeat while temp > 0:
 - Find last digit: digit = temp % 10
 - Increase its frequency: freq[digit]++
 - Remove last digit: temp = temp / 10
2. Use a loop from i = 0 to 9.
3. Display freq[i] for every digit.

Default case

Display "Invalid choice".

Step 9: Stop.

Task2: Test case table:

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	Number = 12345, Choice = 1	Unique	Unique	Pass
2	Number = 55262771, Choice = 1	Not Unique	Not Unique	Pass
3	Number = 355263, Choice = 2	freq_5=2, freq_6=1, freq_2=1, freq_3=2, remaining = 0	freq_5=2, freq_6=1, freq_2=1, freq_3=2, remaining = 0	Pass
4	Number = 0	Invalid input	Invalid input	Pass

Task3: C-program:

```
#include <stdio.h>
int main()
{
    int num, temp, choice, digit, i;
    int freq[10] = {0};
    int unique = 1;

    printf("Enter a positive integer: ");
    scanf("%d", &num);

    if (num <= 0)
    {
        printf("Invalid input");
        return 0;
    }
    printf("\nMENU");
    printf("\n1. Check whether all digits are unique");
    printf("\n2. Find frequency of each digit");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);

    temp = num;
    switch (choice)
    {
        case 1:
            while (temp > 0)
            {
                digit = temp % 10;
                freq[digit]++;

                if (freq[digit] > 1)
                {
                    unique = 0;
                    break;
                }
                temp = temp / 10;
            }
            if (unique == 1)
                printf("Unique");
            else
                printf("Not Unique");
            break;
```

case 2:

```
while (temp > 0)
{
    digit = temp % 10;
    freq[digit]++;
    temp = temp / 10;
}
for (i = 0; i <= 9; i++)
{
    printf("\nfreq_%d = %d", i, freq[i]);
}
break;
default:
    printf("Invalid choice");
}
return 0;
}
```

Task4: Output:

```
(kali@kali)-[~/hdoc14.c]
$ ./a.out
Enter a positive integer: 12345

MENU
1. Check whether all digits are unique
2. Find frequency of each digit
Enter your choice: 1
Unique
```

```
(kali@kali)-[~/hdoc14.c]
$ ./a.out
Enter a positive integer: 55262771

MENU
1. Check whether all digits are unique
2. Find frequency of each digit
Enter your choice: 1
Not Unique
```

```
(kali㉿kali)-[~/hdoc14.c]
$ ./a.out
Enter a positive integer: 355263

MENU
1. Check whether all digits are unique
2. Find frequency of each digit
Enter your choice: 2

freq_0 = 0
freq_1 = 0
freq_2 = 1
freq_3 = 2
freq_4 = 0
freq_5 = 2
freq_6 = 1
freq_7 = 0
freq_8 = 0
freq_9 = 0
```

```
(kali㉿kali)-[~/hdoc14.c]
$ ./a.out
Enter a positive integer: 0
Invalid input
```